

What breaks a docking pose

Which ligand properties predict which physical-validity failure, across the PoseBusters benchmark

428 COMPLEXES · 7 DOCKING METHODS · 7,695 POSES

SUMMARY

Docking is scored by whether a predicted pose lands within 2 Å of the crystal structure, but an accurate pose can still be physically impossible. PoseBusters¹ measures this with 18 validity checks and reports failure per *method*. It records nothing about the molecules, so it cannot say which ligands are hard. We computed 27 RDKit descriptors for the 428 benchmark ligands, joined them to all 7,695 published poses, and modelled each check against ligand chemistry.

Each failure mode has its own signature. Strain energy and internal clashes track torsional flexibility; chirality inversion tracks stereocentre count alone; cofactor clashes afflict *small*, stereocentre-rich ligands. The most frequent failure of all — steric clash with the protein — has **no ligand signature**, making it a property of the method rather than the molecule.

Two limits. Only 41 of 126 method × check strata carry enough failures to model, and molecular weight and rotatable-bond count are correlated at $\rho \approx 0.73$ –0.74 throughout, so where both are significant they cannot be separated.

1 What breaks which check

Each check was modelled by logistic regression on seven decorrelated descriptors entered together, fitted separately per docking method, with standard errors clustered on ligand. Table 1 lists every descriptor reaching significance for at least two methods. Odds ratios are per standard deviation with the other six held fixed, so a value below 1 means failure becomes *less* likely as the property rises.

TABLE 1 Ligand properties carrying independent signal, per check. Multivariate models, odds ratio per standard deviation, other descriptors held fixed.

CHECK	LIGAND PROPERTY	DIRECTION	METHODS	ODDS RATIO / SD
energy ratio within threshold	n rotatable bonds	↑ raises failure	4 of 5	2.08–7.90
no clashes with organic cofactors	mw	↓ lowers failure	4 of 5	0.17–0.31
no clashes with organic cofactors	n stereocentres	↑ raises failure	4 of 5	2.28–3.15
no internal clashes	n rotatable bonds	↑ raises failure	4 of 5	1.55–9.53
no volume clash with organic cofactors	mw	↓ lowers failure	4 of 5	0.15–0.35
no clashes with organic cofactors	clogp	↑ raises failure	3 of 5	1.54–2.32
no internal clashes	mw	↑ raises failure	3 of 5	2.70–3.35
bond lengths within bounds	mw	↑ raises failure	2 of 2	3.33–47.06
energy ratio within threshold	mw	↑ raises failure	2 of 5	2.14–3.55
no clashes with organic cofactors	n rotatable bonds	↑ raises failure	2 of 5	1.96–2.29
no clashes with protein	clogp	↓ lowers failure	2 of 6	0.56–0.67
no internal clashes	formal charge	↓ lowers failure	2 of 5	0.73–0.82
no internal clashes	n stereocentres	↑ raises failure	2 of 5	1.66–1.79
no volume clash with organic cofactors	n stereocentres	↑ raises failure	2 of 5	3.11–3.55
no volume clash with protein	clogp	↓ lowers failure	2 of 5	0.57–0.65
sp3 stereochemistry preserved	n stereocentres	↑ raises failure	2 of 3	4.33–10.43

In plain terms, the table says four things.

FLOPPY → STRAINED

The more rotatable bonds a molecule has, the more likely its pose is internally strained or clashing with itself. More torsions means more ways to fold a molecule into itself. This is the most reproducible signal here — it holds for 4 of 5 methods on both checks, and it replicates on held-out data. OR 1.6–9.5 per SD

CHIRAL → FLIPPED

Molecules with many stereocentres get their chirality inverted. If a model places each centre correctly at random, getting all of them right gets rapidly less likely as they accumulate. This is the failure that matters most in practice: an inverted centre is not a distorted molecule but a different compound, often the inactive enantiomer. OR 4.3–10.4 per SD

SMALL → COFACTORS

Small ligands collide with cofactors; large ones do not. The opposite of the intuitive direction, and the best-corroborated result in the table — PoseBusters tests cofactor clash twice, by atom distance and by volume overlap, and the two independent tests agree per method. A likely reading is that a large ligand cannot be placed in a cofactor-occupied pocket at all, while a small one is placed into the contested space. OR 0.15–0.35 per SD

Nothing about the ligand predicts the most common failure of all. Steric clash with the protein fails on roughly 84% of deep-learning poses and was fitted on more methods than any other check, yet only 5 of 42 coefficients reach significance and no property agrees across more than two methods. no signature

Two properties predict essentially nothing, anywhere: **aromatic ring count** (significant in 4 of 41 fitted models) and **halogen count** (2 of 41). If the expectation was that aromatic-heavy or halogenated ligands are systematically harder to dock validly, this data does not support it.

Molecular weight also appears on the internal-geometry checks (3 of 5 methods for self-clash), but less consistently than rotatable-bond count and in the same direction, and the two are correlated — see §2. Smaller effects in Table 1 worth noting: formal charge *reduces* self-clash failure, plausibly because charged groups sit at the ends of a molecule and repel each other rather than folding inward.

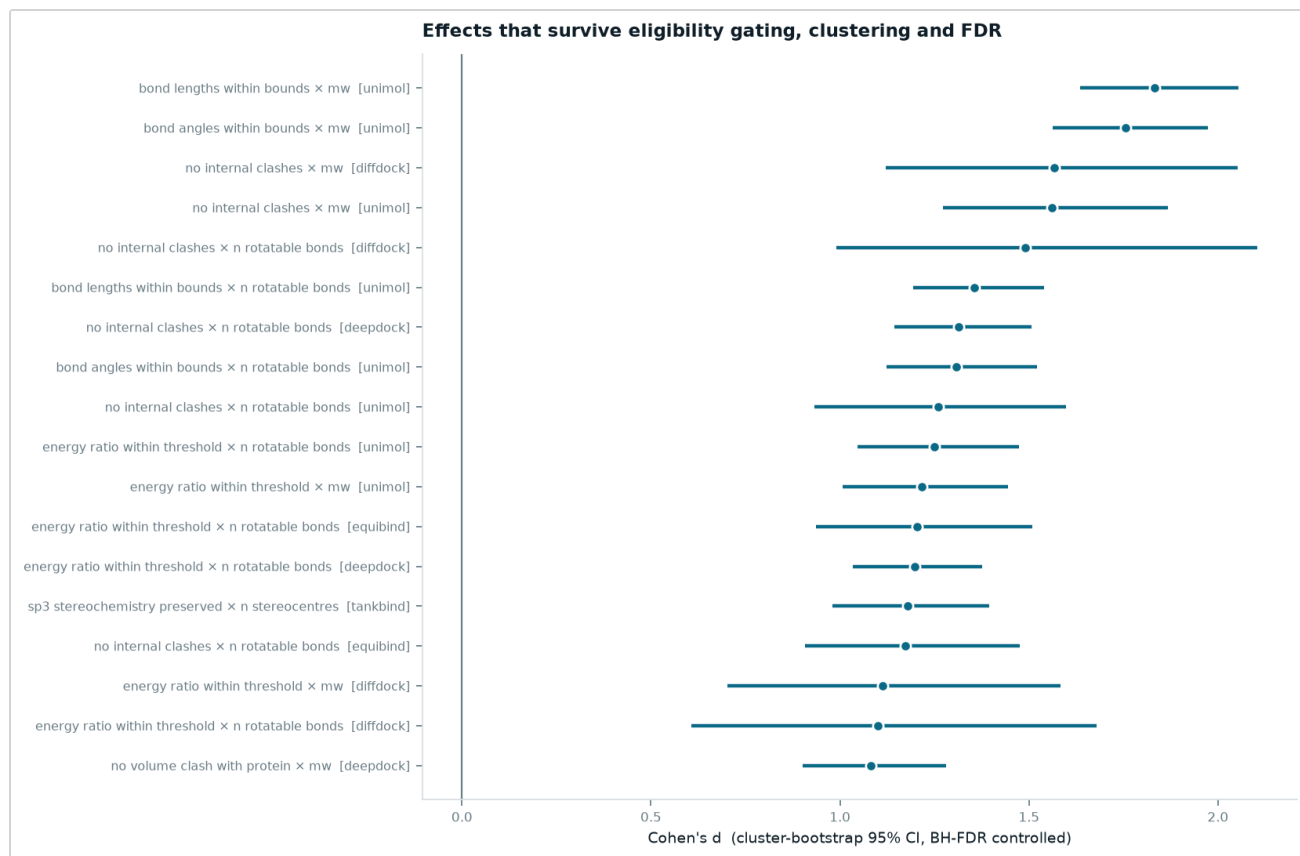


FIGURE 1 The largest single-descriptor associations surviving eligibility gating, ligand-clustered bootstrap intervals and Benjamini–Hochberg correction (95 of 287 estimates survive). Molecular weight and rotatable-bond count recur on the same check for the same method – one signal counted twice, which is why Table 1 uses multivariate models instead.

2 Where size and flexibility cannot be separated

The natural reading of §1 is that torsional freedom rather than bulk drives strain and clash failure. The data supports that only in the weak form stated above. Molecular weight and rotatable-bond count correlate at ρ

≈ 0.73 – 0.74 *within every method's ligand set*, not merely on average, so no stratification of these 428 complexes separates them. For DiffDock — the strongest deep-learning method and the one the question is usually asked about — neither descriptor reaches significance once the other is held fixed ($p = 0.941$ and 0.108), despite both surviving correction individually. Answering the question properly needs a benchmark built to decorrelate the two, with large rigid ligands and small flexible ones in comparable numbers.

3 Crystal contacts are not the explanation

The benchmark's journal version drops 120 of 428 complexes whose ligand touches a crystallographic symmetry mate, on the reasoning that such contacts inflate protein-clash failure. That subset was never published, so we recomputed it from spacegroup expansion: 104 of 427 resolvable complexes have a protein contact within $4.0 \text{ \AA}$ (119 counting solvent), bracketing the authors' 120. Contact status does not explain the clash rate. One method of seven shows a significant difference and it runs the *opposite* way — Uni-Mol's clash failure is 17.1 points *lower* among contact-bearing complexes (95% CI -27.0 to -7.7). The other six straddle zero; Vina and Gold fail so rarely (2 of 323 and 16 of 322 poses) that their point estimates carry no information.

4 How large the gap is

Classical search methods produce accurate poses about half the time and nearly all are valid. Deep-learning methods produce fewer accurate poses, and most of those are invalid.

TABLE 2 Accuracy, validity and their intersection. Raw poses, 428 complexes.

METHOD	CLASS	RMSD $\leq 2 \text{ \AA}$	VALID	BOTH	OF ACCURATE POSES, INVALID
vina	classical	52.3%	97.4%	51.2%	2.2%
gold	classical	51.2%	93.9%	47.7%	6.8%
diffdock	deep learning	37.9%	24.1%	13.6%	64.2%
unimol	deep learning	22.9%	5.6%	1.9%	91.8%
deeppdock	deep learning	17.8%	8.2%	4.7%	73.7%
tankbind	deep learning	15.0%	4.7%	2.6%	82.8%
equibind	deep learning	2.6%	0.9%	0.2%	90.9%

Reporting invalidity only among accurate poses conditions on an outcome that shares an upstream cause with validity. Banding raw RMSD instead uses every pose and needs no such conditioning — and shows the failure is not confined to marginal poses. Deep-learning validity is already poor at sub-Ångström accuracy (DiffDock 61.3%, Uni-Mol 33.3%) and does not recover further out; the classical methods stay above 88% in every band.

TABLE 3 Share of poses passing all applicable checks, by RMSD band. Unconditioned on accuracy.

METHOD	(0.0, 1.0]	(1.0, 2.0]	(2.0, 3.0]	(3.0, 5.0]	(5.0, 10.0]	(10.0, INF]
deeppdock	55.0%	16.1%	12.5%	6.8%	1.1%	0.0%
diffdock	61.3%	20.0%	20.0%	17.8%	18.6%	12.7%
gold	95.2%	88.9%	95.0%	90.7%	97.6%	100.0%
tankbind	36.4%	13.2%	3.0%	1.8%	0.0%	6.9%
unimol	33.3%	2.5%	6.2%	5.2%	4.2%	0.0%
vina	100.0%	93.5%	97.6%	96.2%	96.2%	100.0%

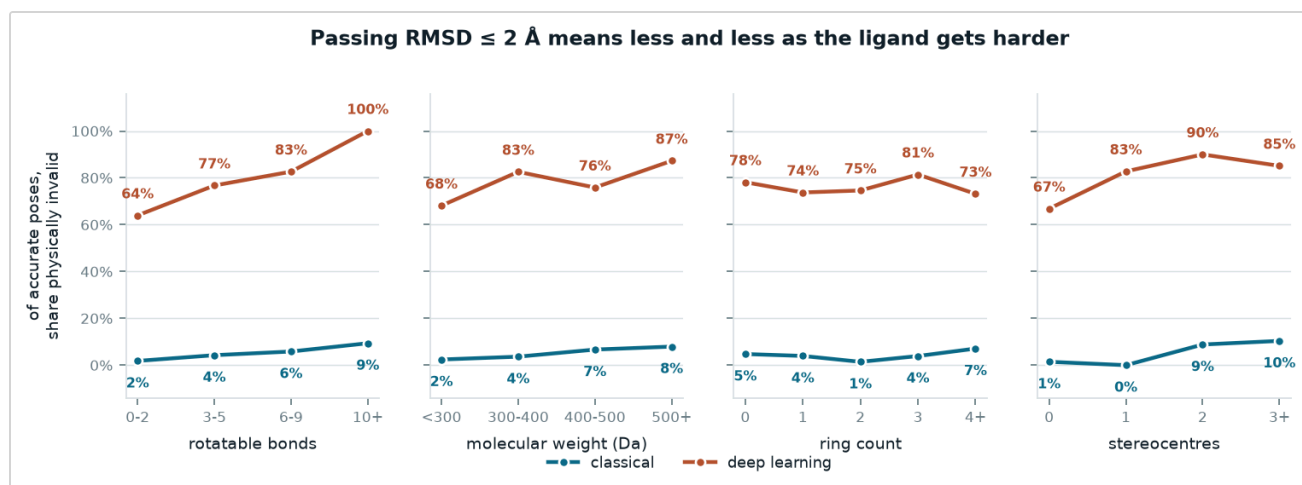


FIGURE 2 Share of RMSD-accurate poses that are physically invalid, across four ligand partitions. Monotone against rotatable-bond count in both method classes, close to flat against ring count.

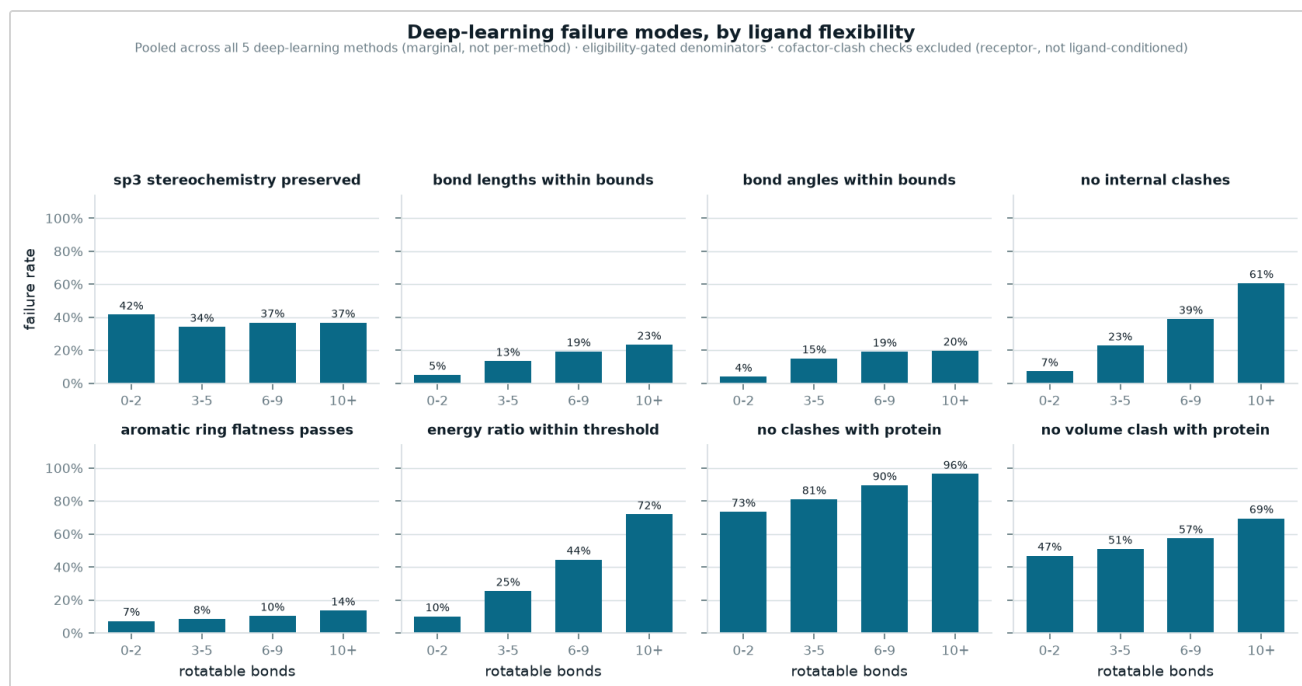


FIGURE 3 Per-check failure against rotatable-bond count, pooled across the five deep-learning methods, with eligibility-gated denominators. Strain and internal clash climb steeply; planarity checks barely move. Cofactor checks are excluded as receptor-conditioned.

5 Confidence in these numbers

The published checks reproduce. Everything here reads the paper's published check outcomes rather than recomputing them, so we re-ran PoseBusters on all 513 crystal ligands against their own receptors: worst-case agreement **99.81%** across 17 checks, with 15 agreeing on all 513 structures. The two disagreements are single structures on different checks.

The main effects replicate held-out. The 85-complex Astex Diverse Set was untouched throughout. Of 14 shortlisted effects, 8 replicate with both intervals excluding zero — the strain and internal-clash effects against rotatable bonds do so for every deep-learning method tested; the chirality and protein-clash effects largely do not.

Two thirds of the grid is unmeasurable. 85 of 126 method $\times$ check strata could not be estimated, 59 of them because the check never failed for that method. Those are not null results; they are absent measurements, and an effect grid without that denominator invites reading silence as evidence.

6 What this means in practice

If you want to know whether a docking method will return a *usable* pose for a particular molecule, three properties carry the information: **rotatable bonds** (predicts strain and self-clash), **stereocentres** (predicts chirality inversion), and **size** (predicts cofactor collisions, in the counterintuitive direction). Flexible, stereocentre-rich ligands are the ones to distrust.

But the single most common failure — steric clash with the protein — is not predictable from the ligand at all. No amount of screening molecules by their properties will avoid it, because it is not a property of the molecule. That one has to be fixed in the method.

7 Method

27 RDKit⁴ descriptors per crystal ligand (size, flexibility, ring system, stereochemistry, polarity, composition), joined to the published results table² on PDB code. Blank check values in that table mean the check was *not run*, not that it failed — both flatness columns are blank for all 2,996 energy-minimised rows — and 85 rows have every check blank because the method produced no pose; treating either as a pass inflates validity.

Each check is restricted to ligands that could fail it: a molecule with no stereocentres cannot fail the chirality check, and counting it as a trivial pass both deflates the failure rate and inflates any effect measured against the gating descriptor. Gating raises the observed chirality-failure rate from 21.4% to 36.7%. The four cofactor checks are further restricted to cofactor-bearing complexes, since they pass almost automatically otherwise (2.4% versus 22.5% failure).

Effects are estimated per method, never pooled. Intervals come from a bootstrap resampling *ligands* rather than poses, since each ligand contributes one pose per method and treating 2,140 rows as independent would report intervals roughly $\sqrt{5}$ too narrow. Strata with fewer than 15 failures or 30 distinct ligands are not estimated. Single-descriptor estimates across the whole grid are corrected by Benjamini–Hochberg⁵ at $\alpha = 0.05$ as one family; because they cannot separate correlated descriptors, the per-check conclusions in §1

come from the multivariate models. Crystal contacts were computed with gemmi⁶ by expanding each deposited structure⁷ by its spacegroup.

Caveats. Descriptors describe the docked molecule, not the returned pose. The predicted poses were never released, so only the crystal-structure rows could be independently recomputed. Several surviving effects rest on small failure counts even after gating — DiffDock's internal-clash effects on 25 failures — so single rows should not be read as decisive.

References

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3. RDKit: open-source cheminformatics. rdkit.org
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Every table is read from the analysis outputs at render time and every figure inlined from its original, so this document cannot diverge from the analysis it reports. Deterministic under fixed seeds; 37 tests. Regenerate with `python -m pb.paper_html`.